RISHII KUMAR SINGH

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SUMMARY

Environmental Engineering undergraduate at DTU researching computer vision and NLP at AIMS Lab, VAIL and UMTRI, with a first-author paper accepted at the WVAQ Workshop at WACV 2027. Builds Python tools and vision pipelines.

EDUCATION

Delhi Technological University
B.Tech, Environmental Engineering | First-year CGPA: 8.35/10 *2025 - 2029 (expected)*

EXPERIENCE

Research Intern, *AIMS Lab, DTU | Prof. Dinesh K. Vishwakarma* Jun 2026 - Present
- Benchmarked 73 image-quality metrics against 5 human references, incl. a self-run 20-rater study (TrustUIQA).
- Led restoration and the embedded model for the AIMS-DTU team, 1st of 9 teams at MaCVi 2027 EURS.

Researcher, *Vision and Interpretability Lab (VAIL) | Remote* Oct 2026 - Present
- Leading real-world data collection and ground-truth documentation for a computer vision dataset and benchmark.
- Contributing to a second computer vision project in a lab focused on vision, interpretability and AI safety.

Remote Research Collaborator, *UMTRI, University of Michigan | Dr. Arpan Kusari* Sep 2026 - Present
- Leading the next version of the MatPredict benchmark for material segmentation and inverse rendering.
- Reproduced v1 object-disjoint segmentation and inverse-rendering results; built bootstrap-CI evaluation tooling.

AI/ML Intern, *WESEE, Indian Navy | Ministry of Defence* Jun 2026 - Jul 2026
- Fine-tuned computer-vision models for satellite-image object detection and segmentation using Python and PyTorch.
- Reviewed data duplicates and split leakage, and supported active learning to improve evaluation-data quality.

AI & ML Intern, *Artha Research & Intelligence Lab | Remote* Oct 2025 - Dec 2025
- Preprocessed 488 facility records across 12 districts using Python and pandas for healthcare-accessibility analysis.
- Mapped facilities, settlements and road closures with Folium to visualize access context in Himachal Pradesh.

PUBLICATIONS

- **R. K. Singh**, A. Rajput, D. K. Vishwakarma. *Do Underwater Image Quality Metrics Agree with Human Perception? A Benchmark with Human Choices, Simple Edits and Generative Drift.* WVAQ Workshop at WACV 2027 (accepted).
- *5th Workshop on Maritime Computer Vision (MaCVi): Challenge Overview.* WACV 2027 Workshops (accepted). Included as an author from the winning team.
- Multilingual knowledge-graph completion paper. Under review, ECIR 2027. Co-author.

PROJECTS

Resilience Graph AI | *ET AI Hackathon 2026, team of 4* | GitHub
- Built detection, ATT&CK retrieval and next-technique prediction for a FastAPI/React cyber-defense prototype.
- Compared autoencoder and IsolationForest detectors on a CIC-IDS2017 day split: PR-AUC 0.570 vs 0.473.

VoxSplit | *Speech-separation research prototype, team project* | GitHub
- Fine-tuned SepFormer with one-and-rest PIT for recursive speech separation, plus a log-mel CNN count/stop model.
- Recorded 71.25% exact speaker count (57/80 custom mixtures); scored separation only on count-correct cases.

MERGED OPEN-SOURCE CONTRIBUTIONS

- **Heliox-OS:** 10 PRs incl. WebSocket token streaming; upstream tests in scikit-bio #2556 and scikit-verify #32.

TECHNICAL SKILLS

Languages & Web: Python, C/C++, JavaScript, TypeScript, SQL, FastAPI, Flask, React, PostgreSQL
ML & Tools: PyTorch, JAX, scikit-learn, NumPy, pandas, OpenCV, ONNX, TensorRT, Git, pytest, Docker

ACHIEVEMENTS AND LEADERSHIP

- **Hackathons:** Winner, BHASHINI VYOMA 2026; Green Tag, BITS APOGEE 2026; 3rd place, BITCAMP 2026.
- **GSSoC 2026:** Ranked 1,662 of 47,951 contributors (top 3.5%); 13 merged PRs across 3 projects.
- **AIMS-DTU:** Co-Head of the university's AI student society, Sep 2026 - Present.